worksheet

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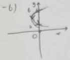
9.1

1. (a): $x = t^2 + 4t \quad t = 2 - y \rightarrow x = 4 - 4y + y^2 + 8 - 4y \rightarrow x = y^2 - 8y + 12$

$x = (y-2)(y-6)$ $\quad t = 4 \rightarrow (0, 6)$

$t = 0 \rightarrow (0, 2)$



(b): $x = (y-2)(y-6)$

2. (a): $\cancel{x^2 = 4\cos^2\theta \quad y^2 = 1 + 2\sin\theta + \sin^2\theta \rightarrow 4\cos^2\theta + \sin^2\theta + 2\sin\theta + 1 = 1}$

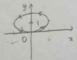
$\rightarrow \cancel{4 - 4\sin^2\theta + \sin^2\theta + 2\sin\theta = 0} \quad \cancel{-3\sin^2\theta + 2\sin\theta + 4 = 0} \rightarrow \sin\theta =$

3   2

-1   2

$\frac{x}{2} = \cos\theta, \; y - 1 = \sin\theta \rightarrow \frac{x^2}{4} + (y-1)^2 = 1$

(b):



$\theta = 0 \rightarrow (2, 1)$

$\theta = \frac{\pi}{2} \rightarrow (0, 2)$

9.2

1. $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \rightarrow \frac{dx}{dt} = 2e^{2t}, \quad \frac{dy}{dt} = e^t\sin t + e^t\cos t = e^t(\sin t + \cos t)$

$\rightarrow \frac{dy}{dx} = \frac{\sin t + \cos t}{2e^t} \quad \frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \Big/ \frac{dx}{dt} = \frac{2e^t(\cos t - \sin t) - 2e^t(\sin t + \cos t)}{4e^{2t}} \Big/ 2e^{2t} = -\frac{\sin t}{2e^{3t}}$

2. (a): $\frac{dy}{dx}$ 가 수평 $\rightarrow dy = 0 \quad dx \neq 0 \quad \frac{dx}{dt} = -2t \quad \frac{dy}{dt} = 1 - 3t^2 \rightarrow t = \pm\frac{1}{\sqrt{3}}$

$t = \frac{1}{\sqrt{3}} \rightarrow x = \frac{2}{3} \quad y = \frac{2}{3\sqrt{3}} \quad \left(\frac{2}{3}, \frac{2}{3\sqrt{3}}\right) \rightarrow \frac{4}{9} + \frac{4}{27} = \frac{16}{27}$

(b): $\frac{dy}{dx} = \frac{3t^2 - 1}{2t} \quad C: \frac{3t^2-1}{2t}(x - 1 + t^2) + t - t^3 \rightarrow 2\int_0^1 \frac{3t^2-1}{2t}(1 - t^2 - t) + t - t^3 \, dt$

3. $\frac{dx}{dt} = \cos t + (-t\sin t) \quad \frac{dy}{dt} = \sin t + t\cos t \quad \frac{dy}{dx} = \pi \quad y = \pi(x + \pi)$

4. $\frac{dx}{dt} = 6t \quad \frac{dy}{dt} = 6t^2, \quad \int_0^2 \sqrt{36t^2 + 36t^4} \, dt \rightarrow \int_0^2 6t\sqrt{1+t^2} \, dt \quad 1 + t^2 = u \quad 2t = du$

$\rightarrow \int_1^5 3\sqrt{u} \, du \rightarrow [2u\sqrt{u}]_1^5 = 2(5\sqrt{5} - 1)$

5. $S = \int 2\pi y \, dx \quad \frac{dx}{d\theta} = -6\cos^2\theta\sin\theta \rightarrow S = \int_0^{\frac{\pi}{2}} 2\pi \times 2\sin^3\theta \times (-6\cos^2\theta\sin\theta)\, d\theta$

$= -24\pi \int_0^{\frac{\pi}{2}} +\cos^2\theta \sin^4\theta \, d\theta \qquad (\cos\theta\sin\theta)^2 = \frac{\sin^2 2\theta}{4} = \frac{1-\cos4\theta}{8}, \quad \sin^2\theta = \frac{1-\cos2\theta}{2}$

$\rightarrow -24\pi \int_0^{\frac{\pi}{2}} \frac{(1-\cos4\theta)(1-\cos2\theta)}{16} d\theta = -\frac{3}{2}\pi \int_0^{\frac{\pi}{2}} 1-\cos4\theta - \cos2\theta + \cos4\theta\cos2\theta \, d\theta$

$\rightarrow \frac{3}{2}\pi \left[\theta - \frac{1}{4}\cos4\theta - \frac{1}{2}\cos2\theta\right]_0^{\frac{\pi}{2}} - \frac{3}{2}\pi \int_0^{\frac{\pi}{2}} 2\cos^3 2\theta - \cos2\theta \, d\theta$

$\cos^2\theta = \frac{1+\cos2\theta}{2}$

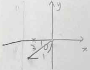
$2\cos^2\theta - 1 = \cos2\theta$

$\cos^2 2\theta = \frac{1+\cos4\theta}{2}$

$2\cos^2 2\theta - 1 = \cos4\theta$

9. 3

1.

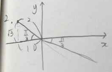


$x = r\cos\theta$

$y = r\sin\theta$

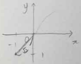
$(1, -\frac{2}{3}\pi) = (-1, \frac{\pi}{3})$

2.



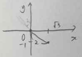
$(-1, \sqrt{3})$

3.



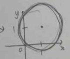
$(\sqrt{2}, \frac{5}{4}\pi), (\sqrt{2}, \frac{3}{4}\pi)$

4.

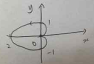


$(-2, \frac{2}{3}\pi), (-2, -\frac{4}{3}\pi)$

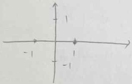
5. $x^2+y^2 = 2r\cos\theta + 2r\sin\theta \rightarrow (x-1)^2+(y-1)^2 = 2 \longrightarrow$



6. (1): $\theta = 0 \rightarrow r = 0, \ \theta = \frac{\pi}{2} \rightarrow r = 1, \ \theta = \pi \rightarrow r = 2, \ \theta = \frac{3}{2}\pi \rightarrow r = 1, \ \theta = 2\pi \rightarrow r = 0$



(2): $\theta = 0 \rightarrow r = 1, \ \theta = \frac{\pi}{2} \rightarrow r = 0, \ \theta = \pi \rightarrow r = -1, \ \theta = \frac{3}{2}\pi \rightarrow r = 0, \ \theta = 2\pi \rightarrow r = 1$



$\theta = \frac{\pi}{6} \rightarrow r = 0$

$\theta = \frac{\pi}{3} \rightarrow r = -1$